

AVANTHI INSTITUTE OF ENGINEERING & TECHNOLOGY

Tamaram, Makavarapalem, Narsipatnam — Visakhapatnam District

AVANTHI CRICKET CARNIVAL

PLAYER AUCTION PORTAL

HACKATHON PROBLEM STATEMENT

A two-phase software engineering challenge

Document	Detail
Version	1.0
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Working window	10 days
Released by	Institution's Innovation Council (IIC), AIET
In association with	Avanthi Cricket Carnival Organising Committee
Audience	Participating student teams

1. The Situation

The Avanathi Cricket Carnival is an institute-level cricket tournament. Eleven franchises, each registered through a faculty coordinator and led by a student captain, build their squads by bidding for student players at a live auction held in the college auditorium.

Earlier editions were run with a simple web page that recorded the auction as it happened. It served its purpose, but it was a scoreboard rather than a system: it did not validate what a franchise could legally spend, it did not enforce squad composition, and it could not recover cleanly from a mistake. This edition is intended to be substantially larger in scale and ambition.

What the new system must solve that the old one did not:

- Bid order — an authoritative record of which franchise bid first, removing any argument in the hall.
- Financial validity — a franchise must be structurally incapable of spending itself into a squad it cannot complete.
- Squad composition — balance requirements enforced continuously rather than checked afterwards.
- Recoverability — a mistyped sale must be reversible at any point, with every dependent figure correcting itself.
- Public transparency — a live, complete record visible to every student, not just the organising table.

Your task is to build the software that runs this event from end to end.

This is not a hypothetical exercise.

The system you design is intended for real use, with real students registering and a real auction conducted in front of an audience. Design accordingly: it must work on the day, under pressure, with no developer standing by to fix it.

2. Glossary

Term	Meaning
Franchise	One of eleven teams. Registered and accountable through its faculty coordinator, with a student captain who leads the squad on and off the field.
Purse	The credits a franchise has available to spend. Each franchise starts with 1000.
Base price	The minimum price at which a player enters the auction.
Retained player	The captain and vice-captain, assigned at registration, costing nothing.
Referred player	A student admitted in the current academic year who joined the college through the ACC reference program, assigned to the referring franchise without auction and without cost.
Bucket	A category of player used to enforce squad balance.
Lot	One player's appearance at the auction.
Super Admin	The tournament organiser. Full control over the system.

3. Actors

Actor	Access
Super Admin	Full control over everything. One account.

Actor	Access
Operator	A second staff account that can run the auction alongside the Super Admin, but cannot change tournament settings or delete franchises.
Franchise	Registers the team through its faculty coordinator, selects captain and vice-captain, declares referred players, bids during the auction. Eleven accounts.
Player	Registers and maintains their own profile. Up to 500 accounts.
Public	No login. Read-only access to everything except phone numbers.

4. Academic Structure

Players are drawn from three courses:

Course	Program	Branches	Years
UG	B.Tech	ECE, EEE, ME, CSE, CSM (AI & ML), CSD (Data Science)	1 – 4
Diploma	Polytechnic	CM, EC, EE, M	1 – 3
PG	M.Tech, MBA, MCA	various	1 – 2

4.1 Roll Number Parsing — a required feature

A student's year of study must never be typed or selected. It must be derived from the roll number, because the roll number is authoritative and self-reported years are not.

B.Tech regular — YY811Abbnn

- YY = year of admission; 811 = college code; 1A = regular B.Tech
- bb = branch code: 02 EEE, 03 ME, 04 ECE, 05 CSE, 42 CSM, 44 CSD
- Year of study = (current academic year – YY) + 1

B.Tech lateral entry — YY815Abbnn

- Identical, except the fifth character is 5 instead of 1.
- Lateral entrants join directly in second year, so year of study = (current academic year – YY) + 2

Worked example — read this twice

25811A0403 is a second-year ECE student. 25815A0403 is a THIRD-year ECE student — same admission year, but lateral entry. Getting this wrong misclassifies the player into the wrong bucket and corrupts every squad rule downstream.

Diploma — YY597-BB-nnn

- 597 = institution code; BB = branch: CM, EC, EE, M
- There is no lateral entry in diploma.
- Year of study = (current academic year – YY) + 1

Example roll numbers: 24597-CM-015, 24597-EC-032, 24597-EE-005, 24597-M-041

Postgraduate

The roll number is recorded, but program and specialisation are selected by the student and verified manually. PG players carry no squad requirement, so nothing structural depends on parsing them.

Academic year rollover

1 July. From that date every student advances one year.

Detained students

A small number of students repeat a year, so their actual year differs from what the roll number implies. The system must allow the Super Admin to override a player's year manually, with the correction logged. A student should also be able to flag the discrepancy during registration so that it reaches the Super Admin as a queue item, rather than being discovered mid-auction.

4.2 Buckets

Five buckets govern squad balance:

Bucket	Contents
B1	B.Tech Year 1
B2	B.Tech Year 2
B3	B.Tech Year 3
B4	B.Tech Year 4
B5	Diploma — all three years together

PG players belong to no bucket and carry no requirement.

PHASE 1 — REGISTRATION

5. Player Registration

Registration runs from 1 to 10 October. Expect up to 500 registrations, the overwhelming majority submitted from mobile phones.

Required of every player

- **Roll number** — the unique user ID. One registration per roll number.
- **Mobile number** — unique, and never displayed publicly.
- **Name**.
- **Photograph** — face clearly visible; formal or casual both acceptable. Quality matters: this image is projected on a large screen during the auction. Consider what the uploader must enforce, and how the image should be processed for both large-screen display and fast list loading.
- **Course, program, branch, year** — derived from the roll number per §4.1.
- **CricHeroes profile** — mandatory. See §5.2.
- **Base price**, chosen from a fixed ladder: 20, 30, 40, 50, 60, 70, 80, 90, 100, 120, 140, 160, 180, 200, 230, 250.

5.1 The Skill Profile

This is the single most important part of registration. It is what a franchise studies before deciding how much to bid, and it is what appears on the projector while bidding is live. A flat list of tick boxes is not adequate.

The skill profile is a branching questionnaire. Questions appear only when they are relevant to the answers already given, so that a player who does not bowl is never asked about his bowling action, and a wicket-keeper is never asked to choose between infield and outfield.

Section A — Batting

Question	Options	Shown when
Do you consider yourself a skilled batter?	Yes / No	Always
Batting style	Strike rotator / Aggressive batter / Big hitter	Answered Yes above
Preferred batting position	Opener / Top order / Middle order / Finisher	Answered Yes above
Batting arm	Right / Left	Always — asked of every player regardless of the answer above

Section B — Bowling

Question	Options	Shown when
Do you consider yourself a skilled bowler?	Yes / No	Always
Bowling arm	Right / Left	Answered Yes above
Bowling type	Fast / Spin	Answered Yes above

Question	Options	Shown when
Pace variety	Swing / Seam / Express pace	Bowling type = Fast
Spin variety	Off-spin / Leg-spin / Left-arm orthodox / Left-arm wrist spin	Bowling type = Spin
Bowling role — select all that apply	Powerplay specialist / Economical bowler / Death-over specialist / Wicket-taking bowler	Answered Yes above

Section C — Fielding and Wicket-keeping

Question	Options	Shown when
Are you a wicket-keeper?	Yes / No	Always
Fielding zone	Infield / Outfield	Answered No above
Preferred fielding position	Slip, Point, Cover, Mid-off, Mid-on, Mid-wicket, Square leg, Third man, Fine leg, Long-on, Long-off, Deep mid-wicket	Answered No above

Section D — Experience

Question	Options	Shown when
Highest level played	District or above / Inter-college / School or intra-college / Recreational only	Always
Played in a previous ACC edition?	Yes / No — with team name if Yes	Always

Derived player type

The player never selects an overall category. The system derives it from the answers above and uses it for display and filtering:

Derived type	Condition
Wicket-keeper batter	Wicket-keeper = Yes and skilled batter = Yes
Wicket-keeper	Wicket-keeper = Yes and skilled batter = No
All-rounder	Skilled batter = Yes and skilled bowler = Yes
Batter	Skilled batter = Yes only
Bowler	Skilled bowler = Yes only
Fielder	Neither batting nor bowling claimed

A validation rule you must enforce

A player who answers No to skilled batter, No to skilled bowler and No to wicket-keeper has declared no cricketing skill at all. Registration must not accept this silently. Require at least one of the three, or make the player explicitly confirm that he is registering as a fielder only — franchises need to know which it is before they bid.

5.2 CricHeroes Profile — mandatory

Every player must supply a CricHeroes profile. This is not optional and not merely for verification: franchises need it after the auction to add their purchased players into the team on CricHeroes, where the tournament itself is run and scored.

- **CricHeroes profile URL** — required.
- **CricHeroes-registered mobile number** — required. CricHeroes accounts are keyed to the phone number, and this is what actually allows a franchise to add the player to a team. A URL alone can be mistyped or point at the wrong person. It may differ from the contact number given at \$5, and is equally private.

Do not make this a hard block at the point of submission.

Many newly admitted and casual players will not have a profile on 1 October. If the field simply refuses to submit, those registrations stall and the organiser spends the registration window answering messages about it. Instead allow the player to register with the status "profile creation pending", show a short in-form guide to creating one, and require the Super Admin to resolve the pending status before ticking the player as paid. Nobody is blocked from registering; nobody reaches the auction without a profile.

Career statistics

Self-declared: matches, runs, batting average, strike rate, highest score, wickets, bowling average, economy, best bowling, catches, stumpings.

Note: CricHeroes offers no public API. Do not attempt to scrape it. Statistics are entered by the player and must be publicly labelled as self-declared; the profile link lets the Super Admin and the franchises check those claims for themselves.

ACC reference declaration

Shown to every student admitted in the current academic year: did you join Avanthi through the ACC reference program, and if so, which team referred you?

This is keyed to admission year, not year of study.

A lateral-entry student admitted this year sits in the second year but is a new admission, and may equally have joined through the reference program. Showing the question only to first-years would silently exclude them. Derive it from the admission year in the roll number: the question appears when the admission year equals the current academic year, whether the student is a regular first year, a lateral entrant in second year, or a new PG or diploma admission. A lateral entrant admitted in an earlier year is not a new admission and must not see the question.

Editing

Players may freely edit their own details. The Super Admin can block editing globally or for an individual player.

Payment

The registration fee is collected offline and the fee amount never appears on the site. The Super Admin ticks each player as paid.

Only paid players may be auctioned. No player is ever rejected — unpaid players remain registered and publicly visible, simply not auctionable.

6. Franchise Registration

Each of the eleven franchises registers with:

- Team name (unique) and logo
- Faculty coordinator: name, department, photograph
- **The faculty coordinator's mobile number** — every franchise is registered through it, and it is the account's primary login.

- **The captain's mobile number** — optional second authorised login on the same franchise account, so the captain can operate the bidding screen on auction day without the coordinator having to hold the device. Both numbers are private, and the Super Admin can change either at any time.

Captain and vice-captain

Both must already be registered as players. Once one franchise claims a player, no other franchise can claim him. They cost nothing and sit outside the fifteen auction purchases.

Referred players

Up to five per franchise, restricted to students admitted in the current academic year — which includes lateral entrants sitting in the second year, not first-years alone. They cost nothing, sit outside the fifteen, and never enter the auction. The franchise names them; the player independently names his referring team on his own form. A referred player therefore need not belong to bucket B1; a lateral entrant referred this year sits in B2. This changes nothing about the squad arithmetic, because referred players occupy no auction slot and consume no purse.

Every referral is assigned manually by the Super Admin after checking both declarations against college records — even when the two sides already agree. Where two franchises claim the same player, or the two declarations disagree, the system must surface the conflict rather than silently resolving it.

Purse

Each franchise receives 1000 credits on approval.

7. Squad Rules

Component	Rule
Captain + vice-captain	2 — free, retained
Referred players	0 to 5 — free
Auction purchases	minimum 15; no upper limit beyond the purse
Minimum squad size	17
Maximum squad size	22

Of the fifteen auction purchases, at least two must come from each of the five buckets. That fixes ten of the fifteen; the remaining five are unrestricted and may include PG players.

Think about what this rule demands before the auction even starts.

Eleven franchises each needing two players from every bucket means a minimum number of paid players must exist in each bucket, or the rule becomes impossible to satisfy. Your system should make this visible to the organiser well before auction day, and provide a way to relax a bucket minimum — applied uniformly to all eleven franchises, never to one.

8. Public Access During Phase 1

Anyone, without logging in, can see every registered player with photograph, class, branch, skills, statistics and base price, plus all eleven teams with their coordinators, captains and squads. Phone numbers are the only thing hidden.

PHASE 2 — THE AUCTION

9. Format

The auction is hybrid. The auctioneer calls prices aloud in the hall; franchises register every bid by tapping on their own phones. The software is the authoritative record of who bid and in what order — this is what eliminates the paddle dispute.

The auction may run across multiple sessions over multiple days, with breaks called at any point. State must survive interruption completely.

Four distinct interfaces are needed:

1. Admin control — laptop
2. Projector display — the big screen in the hall
3. Franchise bidding — phone
4. Public live view — any device

10. Draw Order

Buckets are auctioned in this sequence, each exhausted before the next begins:

B.Tech 3rd year → B.Tech 4th year → B.Tech 2nd year → Diploma → B.Tech 1st year → PG (last)

Within a bucket, every player is assigned a random number **scoped to that bucket**. Two selection modes, switchable at any time:

- **Guest mode** — an honorary guest calls a number aloud; the operator enters it and that player comes up.
- **Auto mode** — used when no guest is present; the system draws at random.

No number is ever called twice.

Skipping

The Super Admin may skip a player. Skipped players are recalled at the end of their own bucket at their original base price. If not recalled there, they carry into Round 2.

11. Bidding

- Bidding opens at the player's base price.
- **No jump bidding.** Each tap raises the price exactly one increment: +10 below 100, +20 from 100 to 199, +30 at 200 and above. The 250 ceiling applies to base prices only; live bids may go higher.
- **Timer:** 30 seconds for the first bid, then 20 seconds after each bid, resetting in full every time regardless of how little time remained.
- The timer always runs its full course, even if all eleven franchises have passed — the auctioneer uses that time to talk the room into re-entering.
- **Each franchise's screen offers Bid and Pass. Pass is reversible** at any time before the hammer. The projector and public view must show live which franchises are in play, which have passed, and which are blocked.
- **The sale completes only when the Super Admin presses the hammer.** A timer expiring does not itself sell the player. With no bids, the hammer marks the player unsold.

12. The Hard Problems

These four are the intellectual core of the build. Solving them correctly is worth more than any amount of visual polish.

12.1 Nobody may bid themselves into an impossible position

A franchise must never be permitted to place a bid that would leave it unable to complete its squad. If a team holds 1000 credits and must still buy fifteen players, it cannot spend all 1000 on the first one — it must retain enough to buy the remaining fourteen at their minimum possible cost.

Note that "slots remaining" is not simply fifteen minus players bought. A franchise might hold eleven players yet still owe five mandatory bucket slots — in which case it has five to fill, not four, and must reserve accordingly.

Derive the constraint yourself. The system must enforce it, not warn about it.

12.2 Mandatory slots must remain fillable

A franchise that has one slot left and still needs a diploma player must not be allowed to spend that slot on anyone else. Work out what conditions make a bid illegal, and block it at the point of bidding.

12.3 Scarcity — the subtle one

Consider the following. The diploma bucket holds fourteen available players, the minimum has been relaxed to one per team, and there are eleven franchises. Ten franchises buy two diploma players each. Fourteen players are gone.

The eleventh franchise can no longer meet its requirement, through no fault of its own.

An explicit design decision — read carefully.

It is tempting to solve this by blocking franchises that have already met a bucket minimum from buying more from that bucket. That solution is rejected. No franchise is ever restricted from bidding on a player it can legally afford. A team that wants three diploma players may buy three. The auction is a free market, and the organiser will not distort it to protect a slower team.

Your task is therefore visibility, not restriction.

The system must track, continuously and for every bucket, how many unsold players remain against how many franchises still need one — and surface that to the auctioneer, the projector and the public the moment the remaining supply drops to or below the number of teams still needing it.

The warning is planning information, not an accusation. A franchise that buys three diploma players is playing the auction legitimately — it spotted value and paid for it. Every franchise is expected to plan its purse and its slots, and each is responsible for its own quota. The system's job is to ensure no franchise can claim afterwards that it was not told what the supply looked like.

When a shortfall does materialise, the affected franchise completes its squad through §13 and is never left short. No blame attaches to anyone in either direction.

The hard part is the arithmetic of "still needs one". A franchise needing two diploma players counts differently from one needing a single player, and the count changes after every sale, every undo, and every relaxation of a bucket minimum. Get that wrong and the warning fires at the wrong moment or not at all.

12.4 Undo must be safe

The Super Admin must be able to reverse any sale, at any point in the auction, not merely the most recent. Purse, slots and every limit must correct themselves immediately, and the history of what happened must survive.

What undo is for — and what it is not for.

Undo exists to correct errors in the recording of a sale: the wrong franchise credited, a mistyped price, a lot hammered before a bid registered, a device that failed at the moment of bidding. It is never a means for a

franchise to reverse a purchase it regrets, and a franchise cannot request it. A completed bid is binding. The control belongs to the Super Admin alone and every use is written to the audit log with a reason.

Consider carefully how you store purse and squad counts. A stored total that is updated incrementally will drift after an undo, and you will discover that drift three lots later in front of an audience.

13. Completing Squads — the Endgame

Round 1 runs through all six categories. **Round 2** then reopens every unsold player — including those skipped and not recalled — with every base price reset to 20. Franchises may request that a specific player or an entire bucket be brought up; requests may be submitted before or at any point during Round 2, and the Super Admin approves and orders the queue. Before Round 2 closes, captains may recall players they earlier passed on.

Every franchise must complete fifteen auction purchases and all bucket minimums by the end of Round 2. No franchise is ever cancelled for failing to do so. Instead a cascade applies, under Super Admin control:

Step 1. **Auto-allotment — whenever players remain in the bucket.** Unsold players from the needed bucket are allotted to the franchise automatically at 20 credits each. This is the normal outcome and requires no request from the franchise. Where several franchises need the same bucket and there are too few players to satisfy all of them, the system proposes an allocation — franchise with the most unfilled slots first, ties broken by smallest remaining purse — and the Super Admin approves or overrides it.

Allotted players must be displayed as "Allotted" and never as "Sold", everywhere they appear — squad lists, the projector, the public page and every export. A squad assembled partly by allotment must be honestly distinguishable from one assembled entirely by bidding.

Step 2. **Only when the bucket is genuinely exhausted** — meaning the system certifies that not a single unsold player remains in it — does the Super Admin choose between two remedies:

- **Uniform relaxation** — the bucket minimum is reduced for all eleven franchises equally, never for one.
- **Scouting** — the franchise finds a willing student, who registers and signs at a fixed 20 credits. The Super Admin verifies the roll number against college records and the registration fee is collected as normal. No vote of the other franchises is required: the bucket is empty, the shortage was not of this team's making, and the franchises who bought the remaining players must not hold a veto over the team they disadvantaged.

A franchise may never choose scouting over an available player.

Scouting exists only for genuine exhaustion. So long as any unsold player remains in the needed bucket, auto-allotment applies and the franchise takes that player. A team cannot decline an allotted player in order to bring in an outsider it prefers.

After Round 2, remaining unsold players may be signed directly at 20 credits, first come first served — but only as squad members beyond the mandatory fifteen. The fifteen may only ever be filled through auction proceedings or the cascade above.

14. The Projector Display

Layout, top to bottom:

1. Large player photograph with career statistics alongside
2. Base price and current bid
3. The franchise currently holding the highest bid
4. Prominent countdown timer
5. All eleven franchise logos along the bottom, showing in play / passed / blocked

It must be driven entirely by the system and require no operator attention during a lot. Assume it is being watched by several hundred people from across a hall.

15. Public Live View

Everything the hall sees, with no login: current lot, current bid, highest bidder, timer, who is in play, all squads as they fill, purse remaining for every team, each team's maximum permissible bid, bucket completion status, and full sold and unsold lists.

Nothing is hidden except mobile numbers. The public may see everything and change nothing.

16. Administrative Controls

Hammer; bid on behalf of a franchise whose device has failed; undo any sale, for correction of a recording error only and never at a franchise's request; direct-assign any player to any team at a price the Super Admin types; skip; pause and resume; switch draw mode; relax bucket minimums; export the complete database as a spreadsheet.

Every administrative action must be written to an audit log with the account and timestamp, distinguishing Super Admin from Operator. When a sale is disputed an hour later, "who pressed the hammer" is the first question asked.

17. Cross-Cutting Requirements

- **Mobile-first.** Most registrations and all bidding happen on phones.
- **Reusable across editions.** ACC 2027 must be creatable without disturbing 2026's records.
- **Data safety.** Automatic backups plus manual export. The organiser must always be able to hold the complete dataset as a file.
- **Concurrency.** Eleven franchises may tap Bid within the same second. Decide what happens and make it defensible.
- **Failure tolerance.** Assume a phone will lose signal and the venue's Wi-Fi will wobble. What happens to a bid in flight? What does a franchise see on reconnecting?
- **Security.** Only the eleven franchise accounts may bid. Consider how you prevent a player, a spectator, or a franchise's rival from placing a bid that is not theirs.
- **Privacy.** Mobile numbers must be excluded at the API level, not merely hidden in the interface.

18. Deliverables

1. Working application covering both phases.
2. Database schema with a justification of your key modelling decisions — particularly how you handled §12.4.
3. Short design note (2–3 pages) explaining your solutions to the four problems in §12.
4. Live demonstration: register two players, register a franchise, run a short auction with at least four teams bidding, trigger a blocked bid, and perform an undo.
5. Seed data sufficient to demonstrate the system meaningfully.

19. Evaluation

Criterion	Weight
Correctness of the four hard problems in §12	30%
Completeness against the functional requirements	20%
Data model quality and integrity	15%
Robustness — concurrency, failure, recovery	15%
Auction-day usability: projector and bidding screens	10%
Code quality and documentation	10%

A visually beautiful system that lets a franchise overspend has failed. A plain system that never allows an illegal state has succeeded.

20. Stretch Goals

- Real-time updates without page refresh
- Player search and filtering on the public view
- Squad analysis: batting and bowling balance, spend per bucket
- Auction replay from the audit log
- Printable squad sheets and a tournament-day team card
- Offline tolerance on the bidding screen, with bids queued and reconciled
- Automatic detection of suspicious bidding patterns

21. If You Run Short of Time

Ten days is a demanding window for this scope, and most teams will not finish everything. That is expected. What matters is that you build in the right order, so that what you do finish is coherent and demonstrable.

Build in this sequence. Do not move to a tier until the one above it works.

Tier	Build this	Why
1 — Essential	Player registration with roll number parsing; franchise registration; the data model; a public player list	Without this there is nothing to auction. Roll number parsing drives every squad rule downstream.

Tier	Build this	Why
2 — Core	Auction lot flow, bidding with increments and timer, the hammer, purse tracking, and the two enforcement rules in §12.1 and §12.2	This is the auction. A submission reaching the end of tier 2 is a credible system.
3 — Completing	Undo, the projector display, the public live view, scarcity warnings (§12.3)	These make it usable in a hall rather than at a desk.
4 — Finishing	Round 2, auto-allotment, scouting, audit log, export	Important for the real event, least important for demonstrating that you understood the problem.

A team that finishes tiers 1 and 2 properly will score better than a team that has fragments of all four. Partial work in a later tier earns little if an earlier tier is broken.

Say clearly in your submission which tier you reached and what you left out. An honest account of scope is treated as maturity, not as failure. An undisclosed gap discovered during the demonstration is treated less kindly.

22. Notes for Participants

- Ask questions early. Ambiguity in a specification is normal; discovering it during the demonstration is not.
- The rules in §7, §12 and §13 interact. Read them together before you design your schema — several teams will discover on the last day that their tables cannot express a constraint the rules require.
- Test the unpleasant cases: undo a sale from an hour ago, disconnect a phone mid-bid, bid one rupee over the legal maximum, pass with every team and then re-enter with two seconds remaining.
- The organiser is available for clarification throughout. Use that.

Appendix A — Acceptance Test Cases

These cases define correctness precisely. Your system must produce exactly these results. They describe what the answer is, not how to compute it — deriving the method is your task.

A.1 Maximum permissible bid

#	Situation	Expected maximum bid
1	Purse 1000. No players bought. All five bucket minimums unmet.	720
2	Purse 1000. 14 players bought, all bucket minimums met.	1000
3	Purse 340. 11 players bought, but 5 mandatory bucket slots still unfilled.	260
4	Purse 200. 13 players bought, all bucket minimums met.	180
5	Purse 20. 14 players bought, all bucket minimums met.	20
6	Purse 600. 15 players bought, all bucket minimums met.	600 — no restriction applies

A.2 Bucket eligibility

#	Situation	Expected result
7	Franchise has 1 slot remaining and still needs a diploma player. Bids on a B.Tech 2nd year.	Blocked
8	Franchise has 3 slots remaining and needs 2 diploma players. Bids on a PG player.	Allowed
9	Franchise has 2 slots remaining and needs 2 diploma players. Bids on a PG player.	Blocked
10	Franchise has 20 credits and one unfilled diploma slot. Bids 20 on a diploma player.	Allowed

A.3 Scarcity — warnings, never blocks

#	Situation	Expected result
11	Diploma bucket: 12 unsold, 11 franchises still need one. Franchise A has already met the diploma minimum and bids.	Allowed. No warning yet
12	Diploma bucket: 11 unsold, 11 franchises still need one. Franchise A has already met the diploma minimum and bids.	Allowed — never blocked. Scarcity warning raised on the admin, projector and public views
13	Diploma bucket: 11 unsold, but 6 franchises still need one, two of them needing two players each.	Warning threshold is 8, not 6 — supply counted against players needed, not teams
14	Diploma bucket: 0 unsold, 1 franchise still needs one.	Franchise routed to scouting under §13, with no vote required
15	A sale is undone, returning a diploma player to the pool while a scarcity warning is active.	Warning clears immediately if supply again exceeds need

A.4 Undo

#	Situation	Expected result
16	A sale from 40 lots ago is undone.	Purse refunded, slot freed, player returns to the pool, all limits recalculated, history preserved
17	The undone sale was the franchise's only diploma player.	Diploma minimum becomes unmet again; maximum bid and bucket eligibility update immediately
18	Same sale is undone twice.	Second attempt rejected — no double refund

A.5 Roll number parsing

#	Roll number	Expected interpretation (2026–27)
19	25811A0403	B.Tech, ECE, regular, 2nd year → bucket B2
20	25815A0403	B.Tech, ECE, lateral entry, 3rd year → bucket B3
21	23811A4201	B.Tech, CSM, regular, 4th year → bucket B4
22	24597-CM-015	Diploma, Computer Engineering, 3rd year → bucket B5
23	26597-M-041	Diploma, Mechanical, 1st year → bucket B5
24	26811A0501	B.Tech, CSE, regular, 1st year → bucket B1, reference-program question shown

A.6 Bidding mechanics

#	Situation	Expected result
25	Current price 90. A franchise taps Bid.	New price 100
26	Current price 100. A franchise taps Bid.	New price 120
27	Current price 200. A franchise taps Bid.	New price 230
28	A franchise attempts to bid 150 when the current price is 50.	Rejected — no jump bidding
29	A bid is placed with 2 seconds remaining.	Timer resets to a full 20 seconds
30	All eleven franchises press Pass.	Timer continues to run; any franchise may re-enter
31	Timer expires with a highest bidder, hammer not yet pressed.	No sale recorded — the sale requires the hammer

Teams are encouraged to build these cases into an automated test suite. A submission that passes all thirty-one is very likely to pass §12 of the evaluation.